

PRIMARY CARE: This guidance can be used when a CMO letter advising influenza is circulating has been issued by Scottish Government (exceptions [click here](#)).

SECONDARY CARE: This guidance can be used when influenza is suspected or confirmed at any time.

UNCOMPLICATED INFLUENZA:

- **Previously healthy**
 - NO antiviral treatment is normally indicated
- **At risk group (including pregnant women) but NOT severely immunosuppressed**
 - prescribe oral oseltamivir within 48 hours of onset (or later at clinical discretion)
 - do not wait for laboratory confirmation
- **Severely immunosuppressed** (e.g. patients with haematological malignancy, transplant (HSCT or SOT) or patients being treated with cytotoxic chemotherapy, DMARDs (azathioprine etc), biologics or prednisolone >20mg/day, HIV patient with CD4<200 or <15%) **AND** the dominant circulating strain has:
 - A lower risk of oseltamivir resistance (e.g. A[H3N2] or influenza B) — prescribe oral oseltamivir
 - A higher risk of oseltamivir resistance (e.g. A [H1N1]) — prescribe inhaled zanamivir, or if this is unsuitable or inappropriate, prescribe oral oseltamivir and monitor closely

COMPLICATED INFLUENZA:

REQUIRING HOSPITAL ADMISSION AND/OR SYMPTOMS OF LRTI/CNS INVOLVEMENT OR SIGNIFICANT EXACERBATION OF UNDERLYING MEDICAL CONDITION

- **NOT severely immunosuppressed**
 - FIRST LINE oral oseltamivir (or via NG)
 - SECOND LINE inhaled zanamivir (if unable to use diskhaler seek advice)
- **Severely immunosuppressed** (patients with haematological malignancy, transplant (HSCT or SOT) or patients being treated with cytotoxic chemotherapy , DMARDs (azathioprine etc), biologics or prednisolone >20mg/day, HIV patient with CD4<200 or <15%) **AND** the dominant strain has:
 - Lower risk of oseltamivir resistance (e.g. A[H3N2] or influenza B)
 - FIRST LINE oral oseltamivir (or via NG) If poor clinical response seek advice from ID/Micro.
 - SECOND LINE inhaled zanamivir (or IV)
 - Higher risk of oseltamivir resistance (e.g. A [H1N1])
 - prescribe inhaled zanamivir (or IV)
 - oseltamivir can be used as a bridging treatment until zanamivir is available

DOSAGES: For hospital inpatients state duration and score off administration boxes not required on medicine chart

ORAL OSELTAMIVIR (Tamiflu®) Duration 5 days (consider 10 days if severely immunosuppressed/critically ill) CrCl >60ml/min 75mg TWICE daily

RENAL IMPAIRMENT: Due to clinical experience and good tolerability of oseltamivir the Renal Drug Handbook advises doses which differ from SPC/PHE guidance. These have been used locally for a number of years.

CrCl 31–60 mL/min: 75mg TWICE daily CrCl 11–30 mL/min: 75mg ONCE daily CrCl ≤10ml/min 75mg ONE off dose

DIALYSIS dosages:

HDF	75mg ONCE and then 75mg after every HDF session
HD (used in acute settings e.g. MHDU/SHDU)	30mg ONCE and then 30mg after every HD session
ICU patients on dialysis	Discuss with ICU pharmacist
CAPD/APD	30mg ONE off dose

OBESITY: no dose adjustment

NG OSELTAMIVIR (Tamiflu®) Doses as per oral oseltamivir above. The contents of the capsule pour easily, but are granular in nature. Disperse in water, draw up the entire dose and administer and flush well. Although small particles are visible in the dispersion, this flushes via an 8Fr NG tube without blockage.

INHALED ZANAMIVIR (Relenza®) 10mg TWICE daily for 5 days (up to 10 days (off label) if suspected/confirmed oseltamivir resistance)

Note: this formulation is NOT suitable for use IV or via nebuliser RENAL IMPAIRMENT/DIALYSIS: no dosage adjustment required

IV ZANAMIVIR (Dectova®) – seek advice from ID/Respiratory/Microbiology before use

NOTE: currently only supplied on a named patient basis via compassionate use programme – see flow chart on next page for supply details

For ALL patients calculate [creatinine clearance](#) (CrCl) DO NOT use eGFR CrCl ≥80ml/min: 600mg twice daily (if <50kg seek advice) Total Duration: 5-10 days

RENAL IMPAIRMENT (<50kg seek advice):

CrCl ≥80mL/min	Initial dose: 600mg and 12 hours later, maintenance dose: 600mg BD
CrCl 50-79	Initial dose: 600mg and 12 hours later, maintenance dose: 400mg BD
CrCl 30-49	Initial dose: 600mg and 12 hours later, maintenance dose: 250mg BD
CrCl 15-29	Initial dose: 600mg and 24 hours later, maintenance dose: 150mg BD
CrCl <15	Initial dose: 600mg and 48 hours later, maintenance dose: 60mg BD

DIALYSIS dosages: seek advice

PREGNANCY/BREASTFEEDING: Limited safety data. Oral oseltamivir/inhaled zanamavir can be used in women who are pregnant or breast-feeding when the potential benefit outweighs the risk. Oseltamivir is the preferred drug in women who are breast-feeding.

ANTIBIOTICS IN INFLUENZA POSITIVE PATIENTS: Consider empirical treatment in febrile patients with extensive pneumonia, hypotension, respiratory failure OR in deteriorating patients OR where no improvement with 3-5 days antiviral treatment. Ensure antibiotics are reviewed and stopped in patients who test positive for influenza where there are no signs or symptoms of bacterial co-infection.

Supply of zanamivir aqueous solution for IV use on a named patient basis from GlaxoSmithKline (GSK) – [PHE 2019](#)

Complicated influenza and decision to treat with zanamivir aqueous solution made by the responsible clinical team

Hospital pharmacist or member of clinical team to contact GSK to obtain relevant paperwork.

GSK contact details: 00800 2468 3579 (Freephone) OR

0208 990 4855 (if Freephone is not available) 24 hours per day/ 365 days per year

Paperwork includes: UK Specific Physician's Guidance Document, Patient Medication Request Form and UK Specific Adverse Event Reporting Requirements

Patient Medication Request Form should be completed by the most senior doctor managing the care of the patient in conjunction with the hospital pharmacist wherever possible.

Once the relevant paperwork has been completed, signed and returned, GSK will deliver zanamivir aqueous solution to the appropriate clinical area in accordance with the requirements of the requesting clinical team.

Guidance notes in relation to accessing compassionate use programme

Supply of Zanamivir aqueous solution via the compassionate use programme is only available on a named patient supply basis. Clinicians should therefore make a very careful judgement about the use of unlicensed zanamivir. **The clinician prescribing zanamivir via the compassionate use programme accepts clinical and professional responsibility for their prescribing decision.**

NHS Trusts should also follow their own Unlicensed Medicines Policies and MHRA Guidance Note 14 "*The Supply of Relevant Medicinal Products for Individual Patients*" in conjunction with this guidance. For example (but not limited to), trusts should ensure the recording of batch number and expiry dates of zanamivir aqueous solution received and supplied.

Wherever possible, ordering of zanamivir aqueous solution should follow 'normal' medicines processes in trusts, i.e. the hospital pharmacy should, if possible, order zanamivir aqueous solution.

When a decision to prescribe zanamivir aqueous solution is confirmed, contact should be made with **GSK on 00800 2468 3579 (Freephone) or 0208 990 4855 (if Freephone is not available) – 24 hours per day/365 days per year**, or by email: GSKClinicalSupportHD@gsk.com. (Fax number 0207 192 6397). The hospital pharmacy team can make this initial contact with GSK, or GSK will also accept an initial request from a member of the clinical team looking after the patient (i.e. senior nurse or doctor) [**Note:** Any out-of-hours requests for zanamivir aqueous solution should be notified to the hospital pharmacy team through the usual pharmacy on call/residency arrangements as soon as possible and by close of play the next working day at the latest].

GSK should be contacted directly to confirm hours of operation for dispatch of Zanamivir aqueous solution via the compassionate use programme and arrangements for completion of relevant paperwork, prior to dispatch.

Pharmacovigilance information

Clinical colleagues in conjunction with the hospital pharmacy team **MUST** provide feedback to GSK to aid pharmacovigilance data collection. The case report form and serious adverse event forms provided, should be used for this purpose, as appropriate.